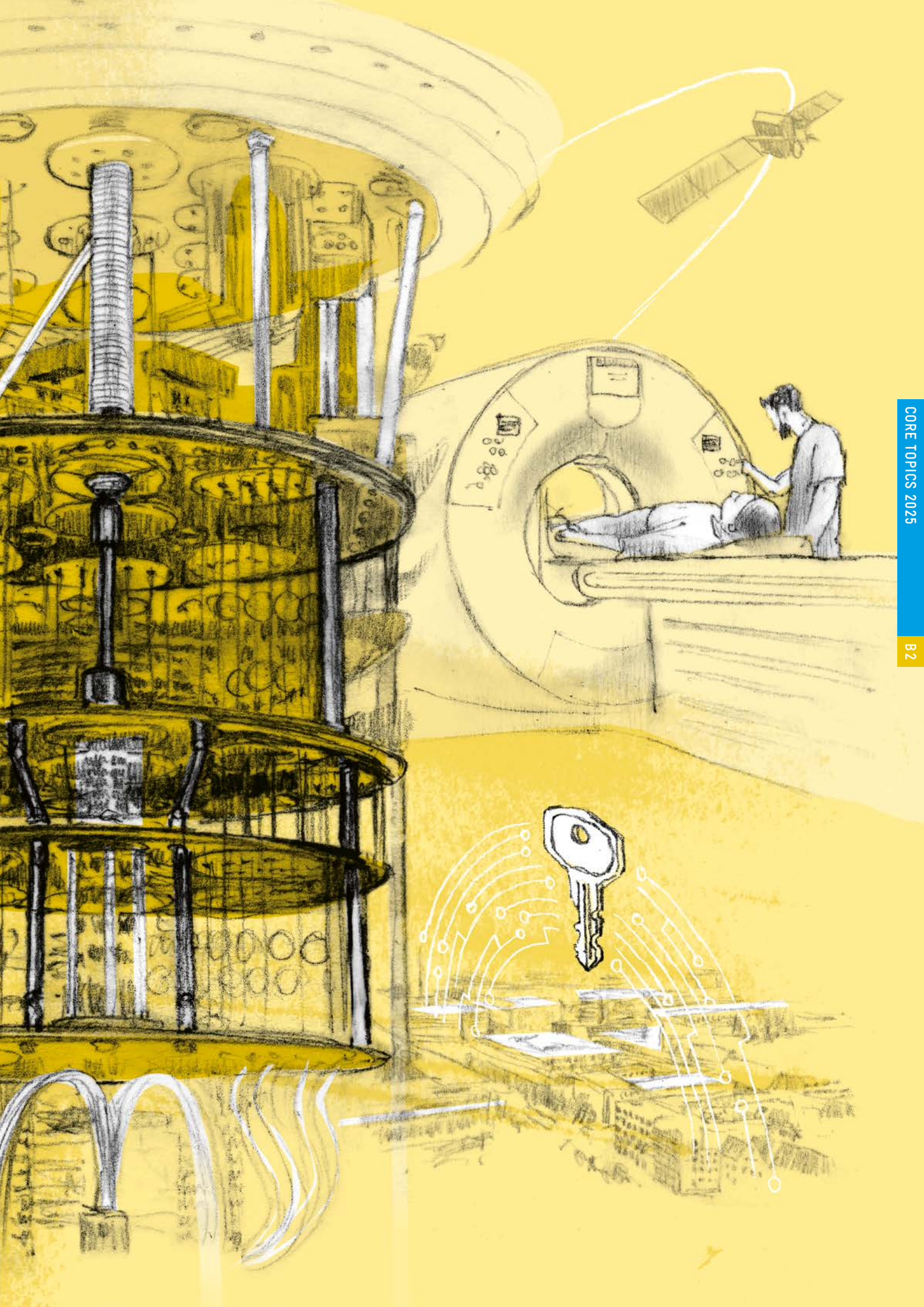


B2 Quantum Technologies



[Download data](#)





B 2 Quantum Technologies

Many cutting-edge technologies, such as computer processors, lasers and satellite navigation, are based on an understanding of the phenomena of quantum physics. New developments in physics now enable the targeted control of quantum systems for the purpose of using quantum effects to develop innovative solutions in existing fields of application, tackle previously unsolvable technical problems and open up new fields of application.

New quantum technologies harbour enormous innovation potential. Quantum computing, for example, promises a significant increase in computing power. Quantum sensor technology significantly improves the sensitivity and precision of measurement and imaging technology. At the same time, these technologies are associated with security policy challenges, as they jeopardize IT and data security. Protection against cyberattacks therefore requires the development of new, long-term secure encryption technologies. Quantum communication offers promising possibilities here.

Many of these new technologies are still in the early stages of development, which offers German players in research and innovation (R&I) the opportunity to actively participate in both this development and the design of future fields of application. Germany is in a strong starting position in this regard, which it must secure in conjunction with its EU partners to remain competitive, particularly vis-à-vis the USA and China. The Federal Government plays a pivotal role here by providing targeted funding for research and development (R&D), creating suitable framework conditions for long-term cooperation between science and industry, and – by means of innovation-orientated procurement – creating future state demand for the output from this

cooperation. The development of an R&I ecosystem should be conceived consistently in European terms. At the national level, the existing regional clusters should be expanded and networked with each other and integrated into a national strategy.

B 2-1 From Quantum Mechanics to Quantum Technologies

125 years ago, German scientists, among others, laid the foundation for a new understanding of physical phenomena by creating the theoretical basis for quantum physics.²⁴¹ German research institutions continue to play a strong role in global quantum research – both in basic physics and in application-orientated R&D.²⁴²

Quantum physics investigates the properties and behaviours of matter, also considering phenomena at the smallest physical level, the so-called ‘nanocosmos’.²⁴³ A cornerstone of quantum physics is quantum mechanics, which has been confirmed by numerous precise measurements (cf. box B 2-1).

High Innovation Potential of New Quantum Technologies

Many of the new quantum technologies are in the early stages of development and are far from market maturity and more widespread use. This assessment is supported by a representative survey conducted on behalf of the Commission of Experts. Only about 4 percent of companies in the information economy and 3 percent in the manufacturing sector say they have looked into quantum technologies intensively or are currently working on or with them (cf. figure B 2-2). A similarly small number of companies

Box B 2-1 Principles of Quantum Mechanics

Quantum mechanics is based on the premise that in so-called **quantum systems** (e.g. an atom), significant changes are triggered by **quantum objects** (individual components of the system, such as an electron) changing their state due to the influence of an externally applied particle (e.g. a photon, a 'light particle').

Quantum mechanics is based on the following principles, which are also key to how quantum technologies work:²⁴⁴

- **Superposition:** In quantum physics, superposition refers to the superimposition of several quantum states. This means that a quantum system is not initially in a clearly defined state without external influence but rather occupies several possible states

simultaneously and the results of physical processes can only ever be predicted in the form of probability.²⁴⁵

- **Decoherence:** Superposition states of a quantum object only exist if the object is completely isolated from its environment. Even the smallest environmental influences cause the superposition to dissolve and thus a specific state to be realized. However, environmental influences are unavoidable in any kind of measurement.
- **Entanglement:** Entanglement creates so-called 'quantum correlations' between particles. As soon as the state of one of two entangled quantum objects is measured in a suitable way and thus determined, the state of the other is also determined, even if they are far apart.

plan to invest in this area over the next five years. In a further question put to the company representatives already familiar with quantum technologies, around 6 percent in the information economy and 4 percent in the manufacturing sector said they intended to invest in quantum technologies either currently or in the next five years.

Quantum technologies are characterized by a high potential for performance improvements. It can also be assumed that they will be used in many industries and in combination with other technologies, thereby unlocking new innovation potential. According to rough calculations, quantum technologies could generate up to US\$2 trillion in added value by 2035 in the chemical, life science, finance and automotive industries alone.²⁴⁶ Since this potential has yet to be realized, quantum technologies are considered to be future key enabling technologies.²⁴⁷

Within quantum technologies, three fields of technology can be distinguished, as described in more detail below: quantum computing, quantum communication and quantum sensing. Transnational patent applications, which are considered to be measurable results of R&D processes, can be used as an indicator of the future relevance of these fields of technology for application in the economy. Transnational patent applications show a high

level of momentum in all three fields of technology. Between 2012 and 2021, the number of patents in quantum computing increased almost fortyfold from a low starting point (cf. figure B 2-3). The number of transnational patent applications in the technology fields of quantum communication and quantum sensor technology has also increased significantly in the past decade.²⁴⁸

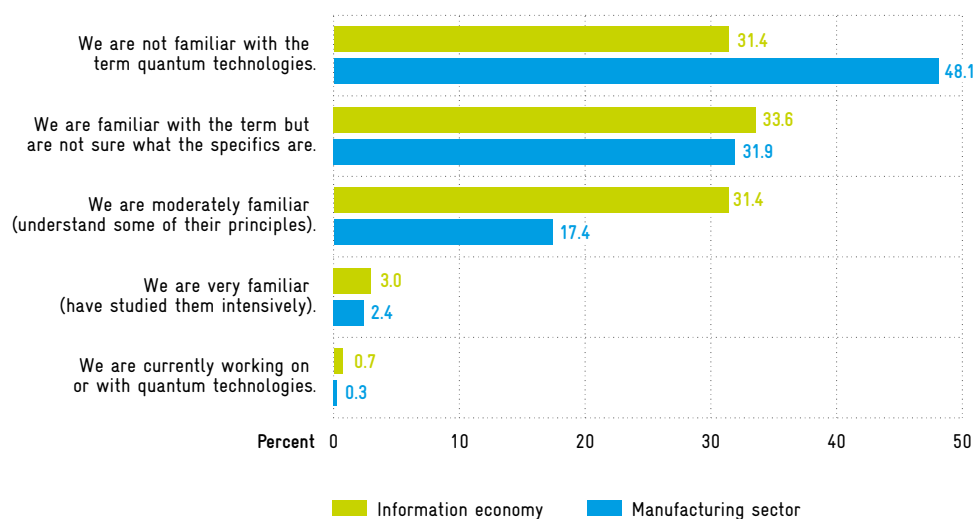
B 2-2 Quantum Computing

Conventional computers have made significant progress in terms of miniaturization and computing power in recent years, but they are increasingly reaching their physical limits. Quantum computing actively exploits quantum phenomena. This could significantly accelerate the solution of highly complex optimization issues in fields such as logistics, pharmaceuticals, capital markets, the environment, health, nutrition, new materials and energy, or even make them possible for the first time.²⁴⁹ In addition, quantum computers could advance key enabling technologies such as artificial intelligence by efficiently processing large amounts of data.²⁵⁰



[Download data](#)

Fig. B2-2 Awareness of quantum technologies in companies in Germany in 2024 in percent

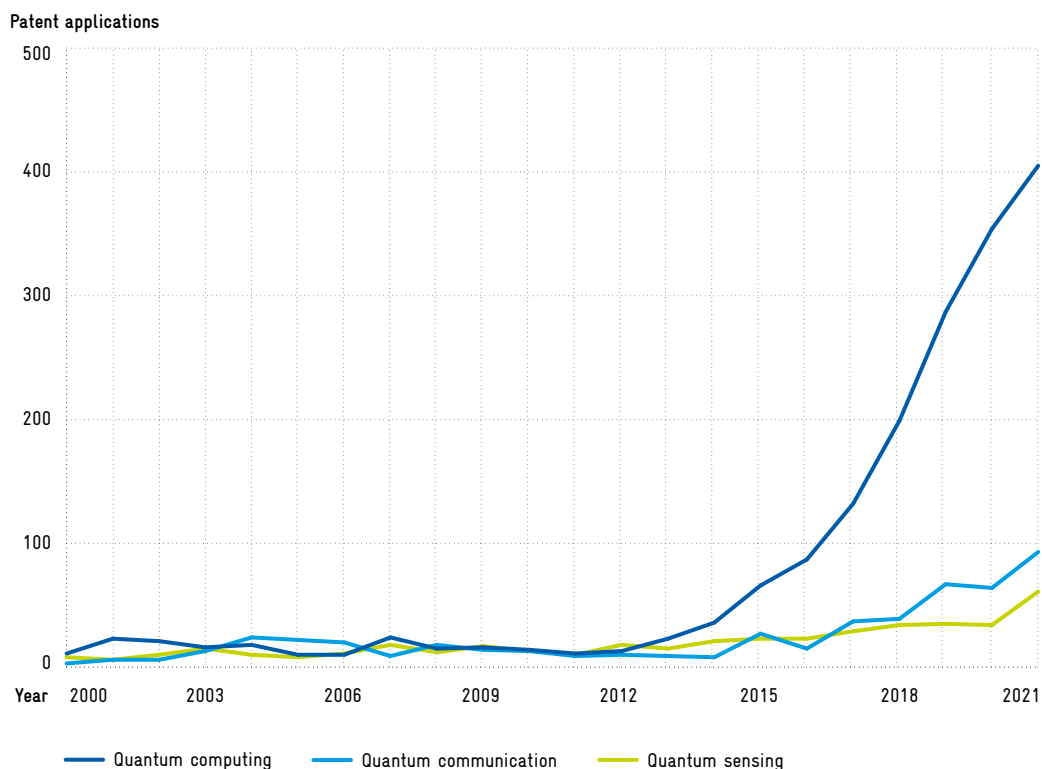


Legend: In the information economy, 0.7 percent of companies stated that they are currently working on or with quantum technologies.
Source: ZEW Konjunkturumfrage Informationswirtschaft Q3 2024.
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Fig. B2-3 Transnational patent applications by technology field worldwide 2000–2021



Legend: In 2021, 405 patents were filed worldwide in the field of quantum computing.
Source: PATSTAT, own calculations.
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Quantum Computers and Hybrid Systems: Immense Potential

Thanks to their special properties, quantum computers can perform many computing operations simultaneously (cf. box B 2-4). This potentially enables them to quickly handle complex calculations that would be impossible or extremely time-consuming for conventional computers.²⁵¹ Examples include the factorization of large numbers, the simulation of complex systems and the optimization of processes. A measurable time advantage over conventional high-performance computers is referred to as a 'quantum advantage' and has already been demonstrated in the context of very specialized calculations.²⁵² In the future, quantum computers will also be able to interact with conventional computers in hybrid form. This involves alternating between the methods best suited to the respective phase and the specific problem in order to solve it.

Quantum computers are based on quantum bits (qubits, cf. box B 2-4), which are based on various so-called hardware platforms.²⁵³ The aim of these hardware platforms is to precisely control as many qubits as possible and use them for calculations. However, the platforms are currently at different stages of development and a dominant technological solution has not yet been established – a typical characteristic of new technologies.²⁵⁴ Depending on the application, different platforms could prove to be particularly suitable in the future. Among the biggest challenges are the limited number of realizable qubits and their high sensitivity to disturbances that can distort calculations. For example, although current NISQ quantum computers (noise intermediate-scale quantum) are able to implement many qubits, their error rate is not yet sufficiently low to allow them to perform calculations reliably.²⁵⁵

Various research activities are currently ongoing in parallel in the field of quantum computing, with different focuses and high potential for added value. Their aim is to develop particularly powerful and compact quantum computers or ones that are compatible with existing systems. These include projects to maximize computing power for applications in various industries,²⁵⁶ for mobile use of quantum computers²⁵⁷ and for integration into existing data centres and supercomputers.²⁵⁸

Given the high technical demands on construction and operation, quantum computing is currently

categorized as data centre technology or high-performance computing.²⁵⁹ Local or cloud-based access options are already being offered as a service to test prototypical applications. However, these applications are currently limited to solving specific optimization issues.²⁶⁰ According to experts, a universal quantum computer that solves a wide range of different problems remains a long-term goal.²⁶¹ It is not yet possible to say whether quantum computers will only be used in data centres in the future or whether, as in the conventional computer industry, there will be a development towards a wide range of devices for use in companies and private households.

One risk of quantum computing is that it is likely to have a significant impact on data security. While traditional encryption methods are currently considered secure, quantum computers could break these in the future, creating new challenges for secure communication (cf. section B 2-3).

USA Overtaken by China in Quantum Computing

The results of a study commissioned by the Commission of Experts show that Chinese institutions have been playing an increasingly important role in research on quantum computing since the early 2000s.²⁶² While around 85 percent of global publications in the early 2000s involved US-American or European participation, China has now overtaken both of these leaders (cf. figure B 2-5).²⁶³ The share of publications with German participation has remained relatively constant over time at 11 to 16 percent. Despite Germany's leading position, no research institution from Germany is among the research institutions with the highest number of publications worldwide in the field of quantum computing. The top 10 consists exclusively of institutions in China, the UK, the USA, Canada and Switzerland. The Max Planck Institute for Quantum Optics is ranked 17th, making it the best-ranked German research institution. This relatively poor ranking is possibly a consequence of the fact that research in Germany is more decentralized than in other countries.

US-American organizations are dominating the patent applications of the past two decades.²⁶⁴ Of the 1,791 transnational patent applications registered worldwide in the field of quantum computing, almost half were filed by the USA (870), followed by the European Union (EU) (282, including 71 from

Box B2-4 Processing Information in Conventional Computers and Quantum Computers

Conventional Computers

Conventional computers store and process information in the form of zeros and ones, known as bits. Put simply, a bit corresponds to the states 'power off' (0) or 'power on' (1). To represent complex information, multiple bits are combined, e.g. into a byte (8 bits), which can represent 256 states in succession. Processing is carried out by transistors arranged in the central processing unit (CPU) – the 'brain' of the system. Modern CPUs with billions of transistors typically work with 32- or 64-bit systems and perform billions of calculations per second. But even supercomputers

reach their limits when it comes to highly complex calculations.

Quantum Computers

Quantum computers store information in quantum bits (qubits), which can assume the states 0 and 1 simultaneously (a state known as superposition, cf. box B2-1). The qubits can form a joint system and thus enable parallel computing operations. With each additional qubit, the number of possible states of the overall system doubles, so that the number of representable states grows exponentially with the number of qubits. However, quantum computers are currently still very prone to failure. For each qubit that provides the computing power, numerous additional qubits are needed for error correction.

Germany), China (159), Canada (118) and Japan (104) (cf. figure B2-5). The leaders in transnational patent applications are the companies IBM, Microsoft and Google, headquartered in the USA, and D-Wave Systems, headquartered in Canada.

A supplementary analysis of company databases identified 473 companies worldwide that intend to commercialize quantum computing.²⁶⁵ They are distributed across the various regions and countries as follows: As with the patent analysis, companies from the USA are strongly represented with 36 percent. China and Germany each account for 7 percent, and the EU as a whole for 19 percent of companies (cf. figure B2-5). Among the identified companies, 131 are start-ups.²⁶⁶ In this group, too, companies from the USA account for the largest share (29 percent).

tum computers can break conventional encryption methods in a short time.²⁶⁷

For this reason, work is being done to make conventional cryptographic methods 'quantum-proof' by using mathematical problems for encryption that are difficult for quantum computers to solve.²⁶⁸ However, the security of such methods depends on the fact that future algorithms or mathematical breakthroughs will likewise be unable to solve these problems efficiently. Quantum communication uses quantum-mechanical principles for encryption, so that security is considered to be independent of future technological developments. It thus enables the interception-proof transmission of messages using so-called quantum key distribution (cf. box B2-6).

Quantum Communication Highly Relevant for Future Security Architectures

The potential ability of quantum computers to decode conventional cryptographic methods means that quantum communication could become an indispensable element of modern security architectures. The security features of quantum communication offer long-term protection for data and messaging, particularly in highly sensitive fields such as political and military communication and the protection of critical infrastructures. For this reason, demand is currently being driven primarily

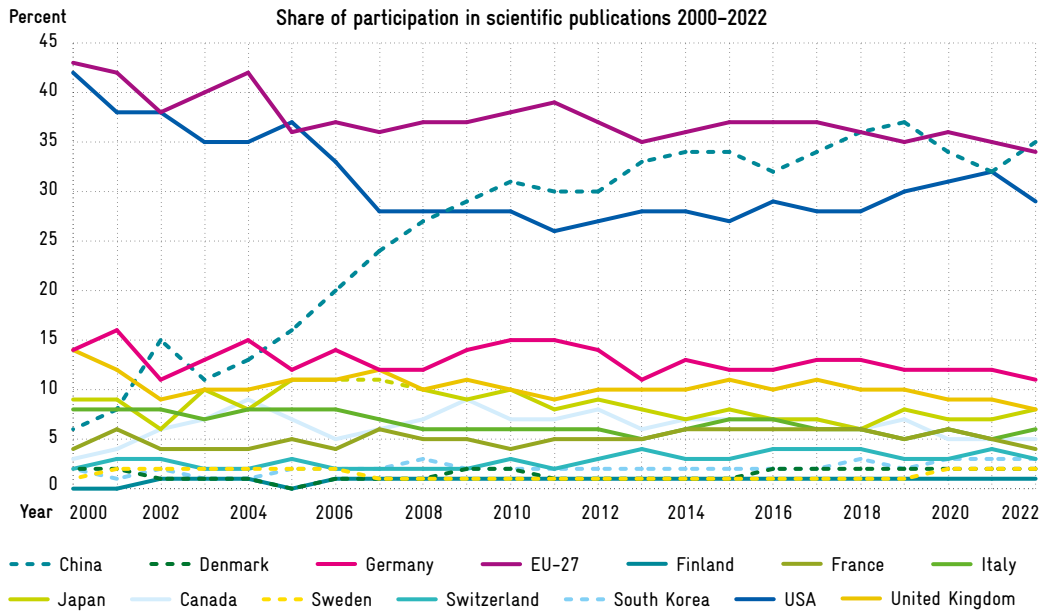
B2-3 Quantum Communication

Encryption ensures the secure transmission and storage of sensitive data. Current encryption techniques are based on mathematical algorithms that are time-consuming for powerful computers to decode, but it is possible in principle. The development of quantum computers could change the dynamics of the race between modern encryption methods and eavesdropping technology, since quan-

Fig. B2-5 Scientific publications, transnational patent applications and active companies in the field of quantum computing

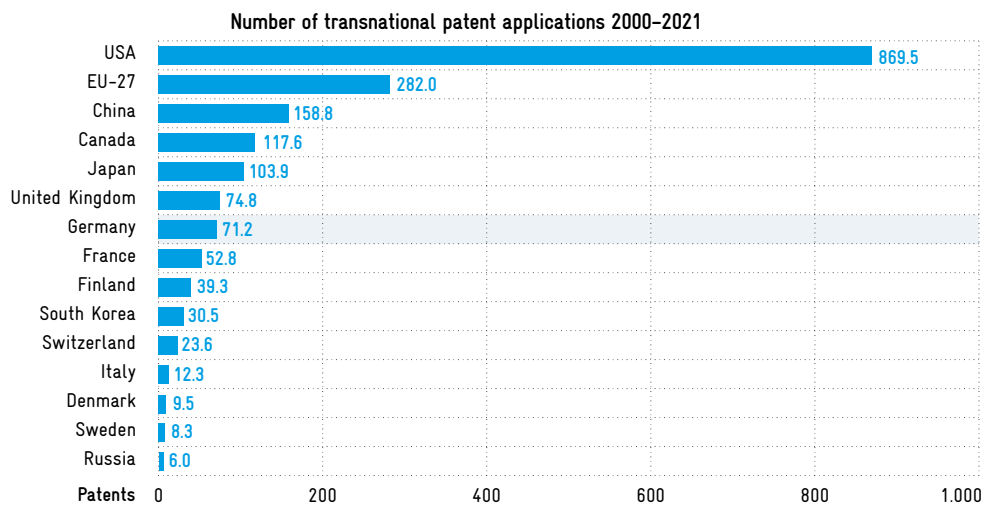


[Download data](#)



Legend: Around 12 percent of scientific publications in the field of quantum computing published in 2020 were written with the participation of authors affiliated with a German organization.

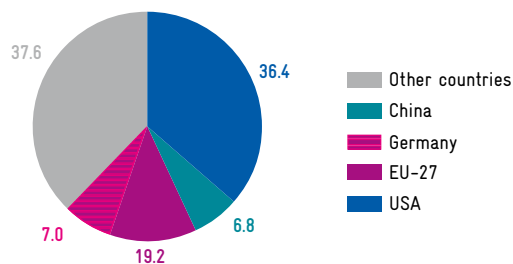
Source: Own representation based on Schmaltz et al. (2025).



Legend: In the period 2000–2021, inventors from Germany accounted for 71 transnational patent applications in the field of quantum computing.

Source: PATSTAT, own calculations. Fractional counting.

Percentage of active companies with a stated focus on quantum computing in 2024



Legend: In 2024, 19.2 percent of the identified companies with a stated focus on quantum computing were based in the EU.

Source: Own representation based on Schmaltz et al. (2025).

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Box B2-6 Quantum Mechanics in Communication: Using Quantum Technologies to Protect against Quantum Technologies

Quantum key distribution (QKD) is a method of securing communication channels. It involves encoding the key for encrypting and decrypting messages in quantum states and exchanging it via specially set up communication channels. The actual message can be transmitted via conventional communication channels. A vivid metaphor for QKD is the exchange of a briefcase containing sensitive information between two diplomats, the contents of which would be destroyed if an unauthorized person tried to access the docu-

ments. In contrast to traditional cryptographic methods, which use mathematical algorithms to generate keys, QKD is based on principles of quantum mechanics (cf. box B2-1). A fundamental concept here is the superposition between two states. It changes measurably as soon as an attempt is made to intercept the information about the state of the quantum system during transmission between sender and recipient. This principle makes QKD a physically secure method for exchanging information. However, the actual security always depends on the specific hardware implementation. Work is therefore being done to develop standardized tests for evaluating QKD hardware.

by public sector players. In the long term, quantum communication could enable the development of a 'quantum network' of quantum computers and sensors.²⁶⁹

Although the first quantum communication methods are already commercially available, major challenges remain. For example, the transmission of information²⁷⁰ is sensitive to environmental influences, which is why the range in existing fibre optic networks is currently limited to local areas.²⁷¹ In addition, the expansion of infrastructure and the miniaturization of hardware require considerable investment.

China and USA Leading in Quantum Communication

China and the USA have been investing large sums in R&D in the field of quantum communication since the 1990s and have made important pioneering contributions. As a result, the participation of Chinese research institutions in scientific publications has increased significantly since the early 2000s. Currently, authors affiliated with Chinese organizations are involved in a significantly larger share of new publications than their colleagues affiliated with EU or US-American organizations (cf. figure B2-7). The share of publications with German participation has remained relatively constant over time at 7 to 13 percent. China's role in the field of quantum communication is underlined by a look at

the research institutions with the highest number of publications worldwide. Of the top 10 research institutions, seven are from China, with the rest coming from the USA, Switzerland and Canada.²⁷²

US-American organizations play a leading role in the transnational patent applications of the last two decades. Of the 526 transnational patent applications worldwide between 2000 and 2021, 135 were filed by the USA. China (98), the EU (94, of which 40 are from Germany), the UK (45) and Japan (38) follow.²⁷³ An analysis of company databases identified 86 companies worldwide that intend to commercialize quantum communication, including 17 start-ups. More than half of these companies are based in China (43 percent) and the USA (16 percent). By contrast, the EU accounts for only 11 percent, with Germany contributing 2 percentage points.

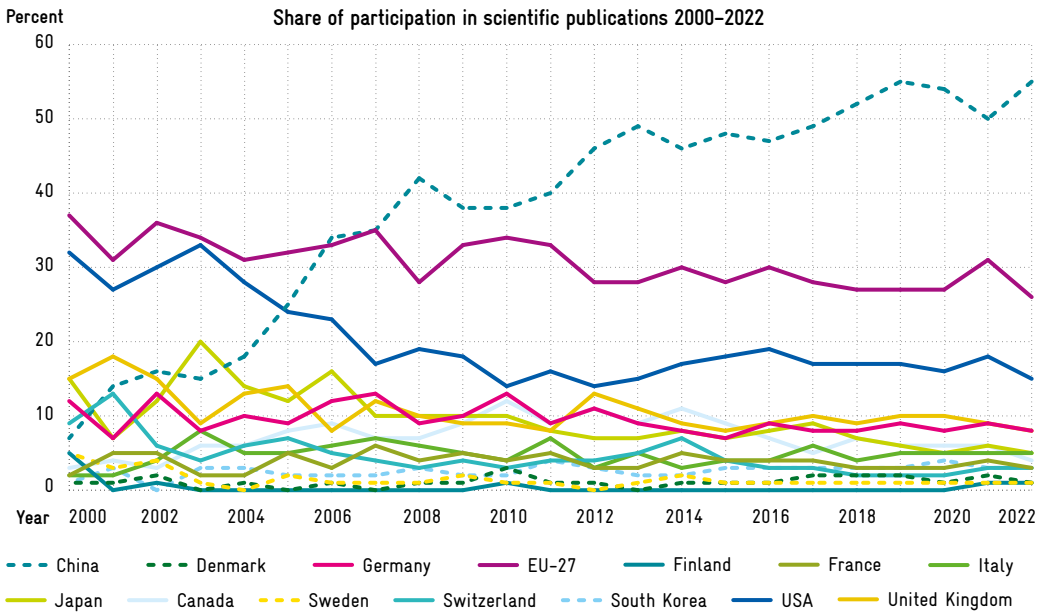
B2-4 Quantum Sensing

Technologies in the field of quantum sensing use the high sensitivity of quantum systems to influences from their direct environment to measure physical quantities (cf. box B2-1).²⁷⁴ This allows environmental variables such as electric and magnetic fields, temperature, pressure or acceleration to be determined with a previously unattainable level of precision. While some sensor technologies based on quantum effects are already firmly established in various markets and it is hard to imagine every-

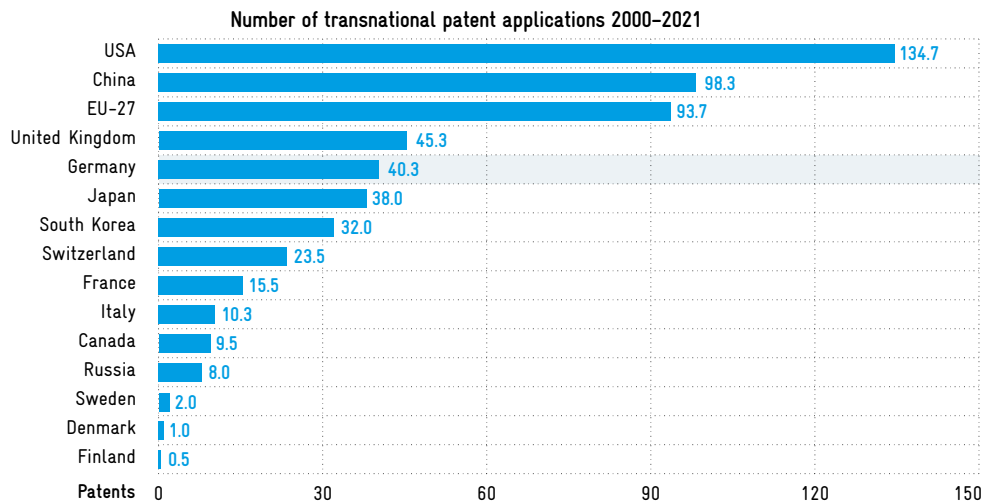
Fig. B2-7 Scientific publications, transnational patent applications and active companies in the field of quantum communication



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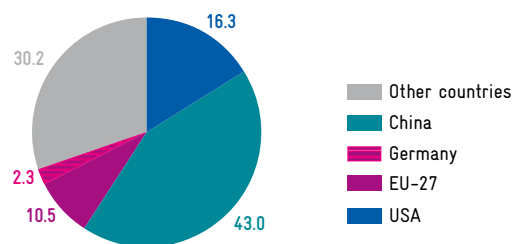


Legend: Around 8 percent of scientific publications in the field of quantum communication published in 2020 were written with the participation of authors affiliated with a German organization.
Source: Own representation based on Schmaltz et al. (2025).



Legend: In the period 2000–2021, inventors from Germany accounted for 40 transnational patent applications in the field of quantum communication.
Source: PATSTAT, own calculations. Fractional counting.

Percentage of active companies with a stated focus on quantum communication in 2024



Legend: In 2024, 10.5 percent of the identified companies with a stated focus on quantum communication were based in the EU.
Source: Own representation based on Schmaltz et al. (2025).
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day life without them, new quantum technologies that enable even greater measurement precision are often still in the early stages of development.

Among the established technologies in quantum sensing, magnetic field sensors are particularly relevant in terms of market volume. They play a key role in medical diagnostics, for example in measuring the weak magnetic fields of the heart or brain using magnetic resonance imaging (MRI). In the form of atomic clocks that send precise time signals for positioning, quantum sensors also provide the basis for today's satellite-based navigation (e.g. GPS). Furthermore, quantum-based technologies are used in semiconductor production.

New Quantum Sensors Enable High-Precision Measurement

Particularly in fields of application with extremely high precision requirements, new quantum sensors can play an increasingly important role. For example, the possibilities for using existing technologies can be significantly expanded by new magnetic field sensors, as they enable more precise measurements or no longer require complex cooling. In addition, the development of optical atomic clocks, which are 100 times more accurate than established atomic clocks, can in future make satellite-based positioning more precise to within a few centimetres or even millimetres. Moreover, novel magnetic field sensors can circumvent the susceptibility of satellite-based systems to interference, thus enabling fully autonomous navigation. Quantum methods with entangled photons promise advances in medical diagnostics, e.g. in the early detection of cancer (cf. box B2-1). Methods such as ghost imaging enable high-resolution images to be obtained with low illumination, for example for imaging at the cellular level. These methods minimize radiation exposure and allow detailed, long-lasting measurements of light-sensitive cells without damaging them.²⁷⁵

Quantum sensors are usually capital goods that are traded on B2B markets and promise excellent value creation potential for applications in downstream markets. The development of new technologies must therefore be closely aligned with the respective user needs. It is expected that advances in quantum sensing will come from continuous improvements rather than radical technological leaps. Particular challenges currently exist in improving user-friendliness, miniaturization and robustness.

USA and China Dominate Publications and Patents

In the field of quantum sensing, Chinese institutions have gained importance in scientific publications between 2000 and 2022 and were involved in 42 percent of the publications in 2022 (cf. figure B2-8). Germany has consistently ranked third behind China and the USA. In the list of the top 10 research institutions with the highest number of publications worldwide, five are from China, two from the USA and one from Germany.²⁷⁶

Between 2000 and 2021, 431 transnational patents were filed in quantum sensing, with the USA, China, Japan and South Korea leading the way (cf. figure B2-8). Germany and France follow at a considerable distance. By far the largest number of patent applications and the three most active patent-filing organizations are in the USA. A review of company databases reveals that only 22 companies worldwide have an explicit focus on quantum sensing.²⁷⁷ The USA (36 percent) and China (32 percent) have larger shares of this market than the EU (18 percent) (cf. figure B2-8).²⁷⁸ It can be assumed that a much larger number of companies, including large corporations, integrate quantum sensing into their applications.

B2-5 EU, USA and China: Comparison of Quantum Strategies

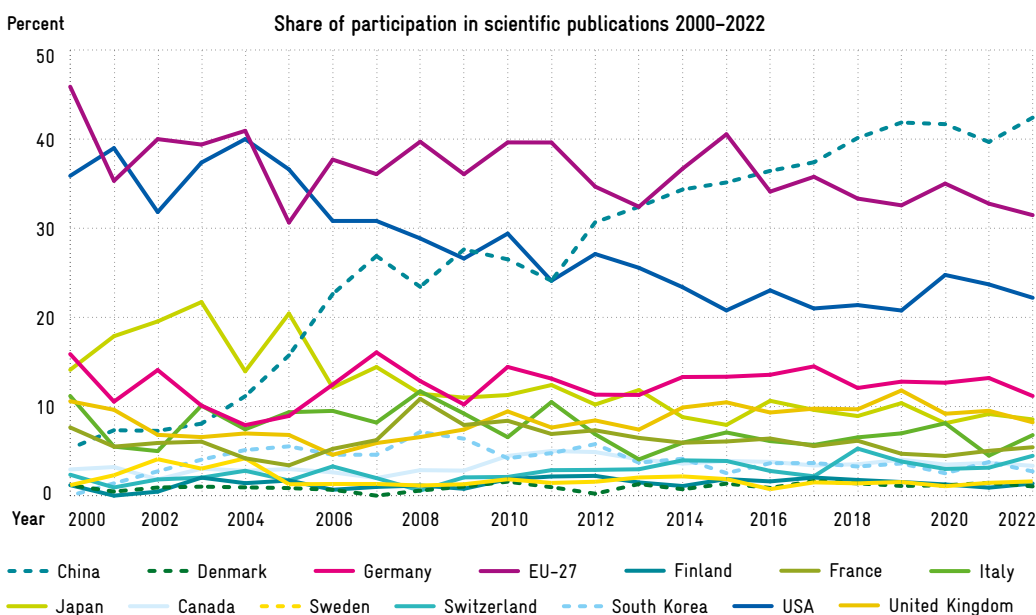
The analyses of publications and patents show that the race for leadership in quantum research is particularly between the US, China and the EU. All three regions are using strategic concepts and targeted support programmes to further develop quantum technologies. These concepts and programmes aim to create quantum ecosystems that coordinate R&I activities in the field of quantum technologies, build a strong infrastructure and provide financial resources.

Funding R&D is one common element. At the same time, the strategies differ in their institutional design and in how they deal with key R&I policy issues. In the USA, R&I is strongly driven by DARPA and the large technology companies. In China, it is predominantly state-controlled. In the EU, the focus is on coordinating national programmes and networking key R&I players.²⁷⁹



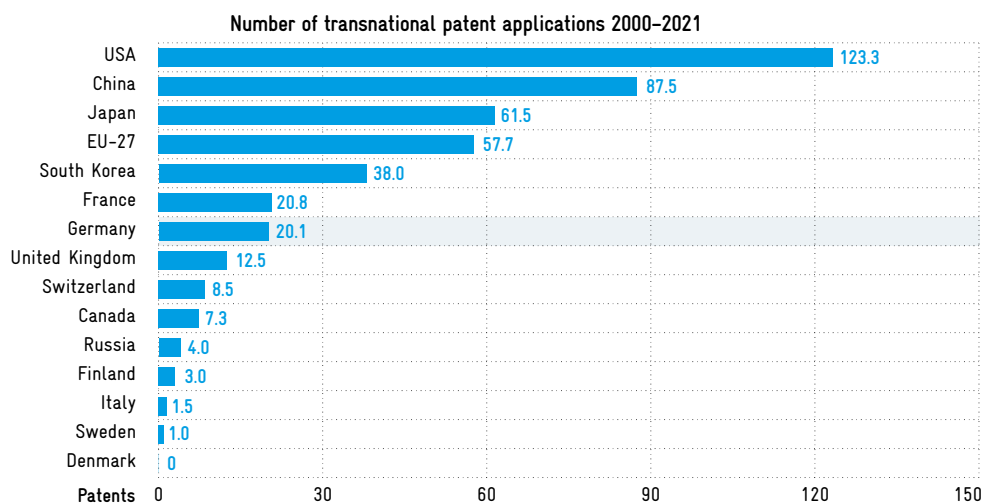
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Fig. B2-8 Scientific publications, transnational patent applications and active companies in the field of quantum sensing



Legend: Around 13 percent of scientific publications in the field of quantum sensing published in 2020 were written with the participation of authors affiliated with a German organization.

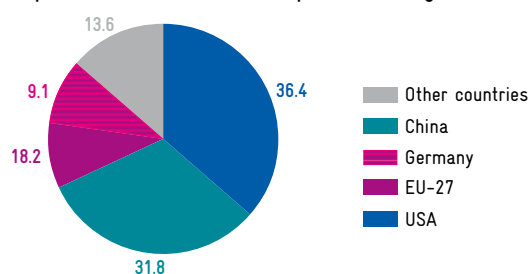
Source: Own representation based on Schmaltz et al. (2025).



Legend: In the period 2000–2021, inventors from Germany accounted for 20 transnational patent applications in the field of quantum sensing.

Source: PATSTAT, own calculations. Fractional counting.

Percentage of active companies with a stated focus on quantum sensing in 2024



Legend: In 2024, 18.2 percent of the identified companies with a stated focus on quantum sensing were based in the EU.

Source: Own representation based on Schmaltz et al. (2025).

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Institutional Framework Varies

In the USA, R&I policy is traditionally mission- and topic-orientated, and there is no independent research ministry. The key players in the strategy launched in late 2018 and coordinated by the National Quantum Initiative (NQI) are the Department of Energy, the Department of Defense and its associated agencies such as DARPA.²⁸⁰ In addition, large technology corporations and private investors in the USA are playing an important role in the further development of quantum computing.

China is pursuing a comprehensive industrial strategy called 'Vision 2035', which is supported by five-year plans and programmes such as 'Made in China 2025'. Quantum technologies, in particular quantum computing and quantum communication, have been a key part of this strategy for 15 years, with the goal of achieving a leading role in important technologies by 2035. Centralized state planning is deeply ingrained, with the Ministry of Industry and Information Technology (MIIT) and the Ministry of Science and Technology (MoST) playing a key role, as does the Chinese Academy of Sciences (CAS), the highest advisory body for science and technology. As the largest research organization in the world, reporting directly to the State Council, the CAS coordinates major initiatives such as satellite projects and the development of quantum communication infrastructures. In 2017, a 'National Laboratory for Quantum Information' was also established, which, according to expert reports, has been provided with substantial funding.²⁸¹

In 2018, the EU launched the Quantum Flagship Initiative, a ten-year plan to promote quantum technologies under the leadership of the European Commission. The focus is on coordinating national programmes, networking key players and promoting strategic partnerships.²⁸² Within the European Commission, the Directorate-General for Communications Networks, Content and Technology (DG CONNECT) has been responsible for digital strategies and the Directorate-General for Research and Innovation (DG RTD) for the funding of projects via Horizon Europe. The European Research Council (ERC) also supports basic research in the field of quantum technologies.

Technology Transfer Included in All Strategies

In the USA, public-private partnerships are accelerating technology transfer, i.e. the transfer of scientific knowledge into economic application. The Department of Energy is working with research institutions and industry to establish five quantum information science research centres (QIS centres) to promote research, prototype development and commercialization.

In China, technology transfer is promoted by the state, with research institutions and companies linked with state support to bring quantum computing to market readiness, for example. Local industry clusters are being established under provincial governments to attract investment and skilled workers.²⁸³

In the EU, the European Innovation Council (EIC) supports innovation and start-ups in the field of quantum technologies through support programmes and financing. The Directorate-General for Internal Market, Industry, Entrepreneurship and SMEs (DG GROW) is promoting the EU's industrial base and competitiveness, including support for SMEs and the promotion of innovation in the field of quantum technologies. The European Investment Bank (EIB) is also providing capital for start-ups and SMEs to finance projects in the field of quantum technologies.

EU Cautious on Civil-Military R&D Cooperation

The interaction between civil and military R&D can lead to efficiency increases in that military R&D contracts awarded to companies trigger further private-sector R&D spending, knowledge and findings from the military sector are adopted in the civil sector (spillover) and technologies can be used for both civil and military purposes (dual use).²⁸⁴ Quantum technologies are relevant for civil and military fields due to their potential range of applications and their security policy significance, for example in quantum communication. Efficiency increases through economies of scale and scope in R&D projects can therefore be particularly effective in quantum research.

This approach is clearly embedded in the strategies and funding initiatives of the USA and China. In the USA, the NQI is regularly supplemented by the National Defense Authorization Act, which, among

other things, sets funding priorities for quantum research with military interests in mind and ensures the coordination of relevant institutions and programmes.²⁸⁵ China has explicitly included the interaction of military and civil R&D, including in quantum research, in the 13th and 14th Five-Year Plan with the aim of strengthening national security and economic competitiveness in equal measure. This is to be achieved by sharing resources, exchanging research results and applying them in both fields.²⁸⁶

The EU has so far been reluctant to do so. Military and civil R&D are strictly separated in the support programmes and interaction between these fields is not described in either the Quantum Flagship Initiative or the broader Horizon Europe programme.²⁸⁷ However, the strategic benefits and possible scenarios for promoting technologies with dual-use potential, including quantum technologies, are also being discussed at the European level.²⁸⁸

Standardization Initiatives Crucial for Progress

Standardization is a key prerequisite for progress also in quantum technologies. Standardized performance tests and benchmarking methods enable the systematic comparison of R&D results.²⁸⁹ In addition, standards promote the interchangeability and interoperability of technical components such as quantum modules, as well as flexible cooperation along the value chain through uniform development environments and software.²⁹⁰

In the USA, the National Institute of Standards and Technology (NIST) is working on new standards for cryptography. It recently published new encryption standards with the participation of German scientists.²⁹¹ In China, the 2024–2027 action plan presented in April 2024 aims to accelerate the development of standards in the field of quantum technologies. In the EU, a technical committee under a German secretariat and co-chair is coordinating the development of relevant standards for quantum technologies and their coordination with international organizations.²⁹²

B 2-6 Germany: National Quantum Strategy and Existing Challenges

With its national quantum strategy ‘The quantum technologies conceptual framework programme’ (Handlungskonzept Quantentechnologien), the

Federal Government has created a funding framework for the period 2023 to 2026 and set out a vision for up to 2036.²⁹³ A total of €3 billion is available for this over four years.²⁹⁴ Table B 2-9 shows that the German concept is very well funded in international comparison.²⁹⁵

The Federal Government’s quantum strategy is open to all technologies, i.e. competing solutions for specific problems within quantum technologies are considered worthy of funding in principle. The funding approach is based on collaborative projects and state co-financing, with public contracts intended to stimulate private investment. In addition to this, the Länder have developed their own funding programmes, from which regional R&I clusters of international renown have emerged, including the Munich Quantum Valley (€80 million, Bavaria) and QuantumBW (€115 million, Baden-Württemberg). Other programmes include the Berlin Quantum Alliance, Hamburg Quantum Innovation Capital and Quantum Valley Lower Saxony.

Future innovations in quantum technologies require significant initial investments in basic research and the necessary infrastructure, while technological uncertainties are high and many applications will not be realizable for several years.²⁹⁶ This can lead to underinvestment by private-sector players, even though quantum technologies offer significant social and economic benefits. From the viewpoint of the players, the Federal Government’s planning horizon up to 2026 is too short to justify long-term investments in R&D in the field of quantum technologies. In view of the long development cycles of the projects, a forward-looking, long-term strategy is needed.

High Demand for Networking Between Players

Within the three technology fields, various technological approaches are being researched in parallel and compared in terms of their suitability for specific applications. One example is the production of qubits in quantum computing, which can be done either on atomic or on solid-state hardware platforms. At the same time, there are close dependencies and interactions between basic research, applied research and application development. For example, a sound understanding of user needs is crucial to drive the further development of hardware and software in a targeted manner. Due to these interdependencies, it is necessary to network



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Tab. B 2-9 Key data of selected quantum strategies

Country	Strategy	Key initiatives	Budget	Term
Germany	Quantum Technologies Conceptual Framework Programme	Quantum Systems Research Programme; IT Security Research Programme; Quantum Computing Initiative	€3 billion	2023–2026
EU	Quantum Flagship	EuroHPC; EuroQCI	€1 billion	2018–2028
France	Quantum Plan – France National Quantum Strategy	–	€1.8 billion	2021–2025
United Kingdom	National Quantum Strategy (NQS)	National Quantum Technologies Programme (NQTP)	NQS: £2.5 billion (ca. 2.97 billion Euro); NQTP: £1 billion	NQS: 2023–2033 NQTP: 2014–2024
China	Made in China 2025; 14th Five-year-Plan	National Key R&D Projects; National Laboratory of Quantum Information	estimated €15 billion	until 2025 2021–2025
Japan	Quantum Technology Innovation Strategy; Vision of Quantum Future Society	Moonshot R&D Program; Quantum Innovation Hubs	259 billion (ca. €1.5 billion)	2018–2023
South Korea	Korea's National Quantum Strategy	–	3 billion (ca. €2 billion)	2023–2035
USA	National Quantum Initiative Act	–	US\$1.2 billion (ca. €1.1 billion)	2019–2023
Canada	Canada's National Quantum Strategy	–	CAS\$360 million (ca. €240.9 million)	2021–2028

Source: Own representation based on Schmaltz et al. (2025).
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R&D activities and to ensure an intensive exchange between the players within a technology field.²⁹⁷

In addition to networking within a technology field, networking between the different fields is equally crucial, since advances in one field, such as quantum sensing, can also advance other fields, such as quantum computing. Many research infrastructures are based on common enabling technologies, such as laser technology, low-noise microelectronics and cryogenics. Good networking makes it possible to identify and exploit the advantages of individual technologies more quickly. In the long term, the integration of the three technology fields of quantum computing, quantum communication and quantum sensing can create a global quantum network (quantum web) where quantum technologies are networked with each other in a non-interceptable manner.²⁹⁸ This requires the early development of standards and specifications, which in turn requires close cooperation between players in all technology fields.

To avoid inefficient parallel structures and to integrate application options into research at an early stage, intensive exchange between the players within a technology field and strong networking between

technology fields are essential. Regional R&I clusters promote the networking of players through geographical concentration, facilitate knowledge spillover and thus strengthen R&D activities.²⁹⁹

Technological Sovereignty in Quantum Technologies at Risk

Technological dependencies on third countries are a risk for Germany in the context of uncertain geopolitical developments. In response to this, German R&I policy has increasingly focused on the goal of technological sovereignty. Long-term technological sovereignty requires forward-looking strategic concepts that aim to drive technological development and standardization forward or ensure that technologies can be obtained from other economic areas in the long term without unilateral dependencies.³⁰⁰

Quantum technologies are undoubtedly among the future technologies for which ensuring technological sovereignty is of vital importance. This also includes mastering the use of quantum technologies by developing and maintaining the necessary knowledge base and skills in Germany. At the same time, it must be ensured that quantum technologies remain available in the long term – either through

independent production within the EU or through a diversified procurement strategy in international trade.³⁰¹ Another approach to avoiding one-sided dependencies is the strategic development of unique selling points. These could be needed by other players in the future and put Germany in a stronger bargaining position in international negotiations ('bargaining chips').

Quantum Computing Associated with Security Risks

Predictions as to when a universally applicable quantum computer will be available for practical applications vary considerably.³⁰² Nevertheless, experts agree that future quantum computers will achieve a level of computing power that will enable them to break current encryption standards in communication and data security. This constitutes a significant risk for the security architectures in military, political and civil communication. Adjustments to the security standards are therefore indispensable to ensure data and communication security in the long term, especially for critical infrastructures. The state has a pivotal role to play here.

The risk of so-called 'harvest now, decrypt later' attacks is particularly critical. In such attacks, sensitive data is intercepted, stored for years or decades and later decrypted with sufficient computing power. In view of the long transition period until the full introduction of new encryption methods, and the fact that these attacks already represent a significant security risk today, the need for action here is urgent.

In addition to the security risks arising from the prospective computing power of quantum computers, risks are foreseeable that could arise from unilateral dependencies in the provision of hardware. For example, a quantum computer that is supplied and operated by a non-European company not only creates dependencies with regard to the availability of the infrastructure but is also problematic for military applications with sensitive data and issues of intellectual property in software and data.³⁰³

Application Limited by Shortage of Skills and of Specialist Workers

In the field of quantum technology, there is a risk that R&D activities will be hampered in the long term by a shortage of highly qualified graduates in

the STEM subjects. According to experts, the limited expertise of the current workforce could also significantly delay the introduction and dissemination of application-ready quantum technologies.³⁰⁴ Germany's (continuing) education offerings in the field of quantum technologies are currently well developed by international standards. In 2021, six German universities offered a master's programme with a specific focus on quantum technologies – more were only available in the United Kingdom (8) and the USA (16).³⁰⁵

In Germany, content on quantum physics and quantum technologies is largely integrated into the curricula of schools and universities.³⁰⁶ In addition, targeted support measures have been launched³⁰⁷ to develop expertise and train young talent in the field of quantum technologies, for example through the Quantum Futur³⁰⁸ young talent development programme and the introduction of part-time courses of study.³⁰⁹ However, it remains uncertain whether these measures are sufficient to meet specific recruitment needs in the long term.³¹⁰

Commercialization of Quantum Computing Challenging

One of the characteristics of a high-performing innovation ecosystem is its ability to translate insights from basic and applied research into innovative, marketable products.³¹¹ Figure B 2-10 shows that researchers from the EU have been involved in a large proportion of global publications on quantum computing over the past two decades, with 12,121 publications (green bars).³¹² However, only just under 10 percent of publications with EU participation have already been cited in patents granted by the end of 2023, while this figure is almost 20 percent for publications with US-American participation, twice as high (blue bars). In China, the figure is much lower at 5.5 percent.³¹³

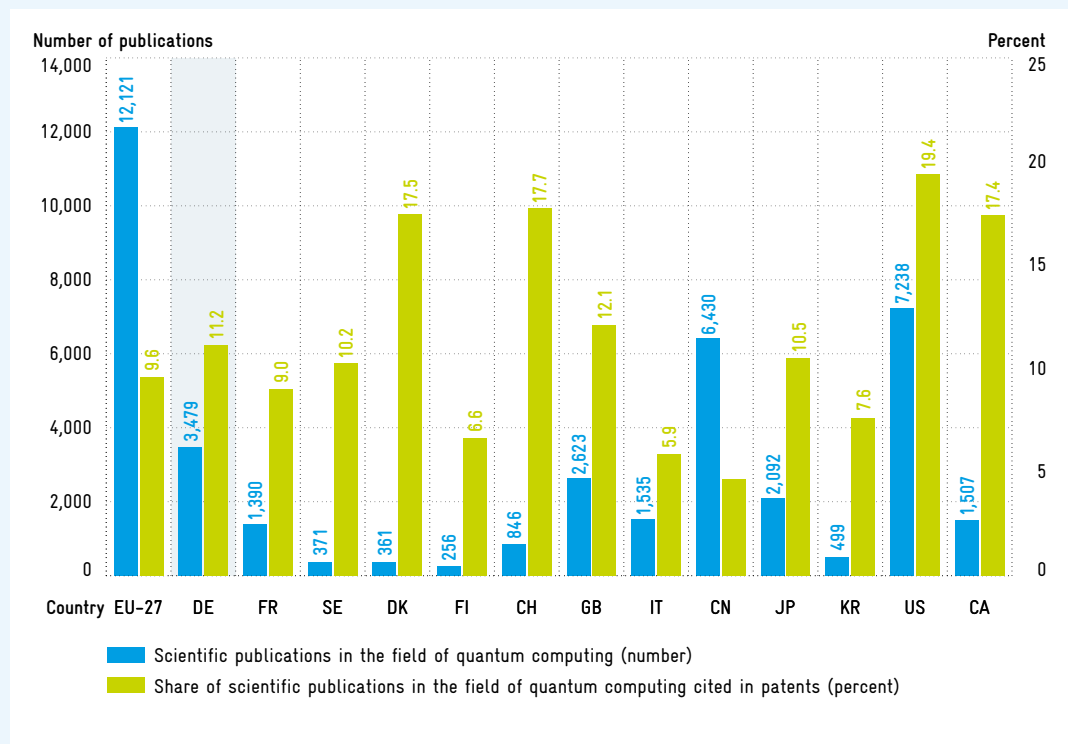
Weaker networks between research and application in the EU, differences in the type of research, less career mobility, more red tape and fewer incentives to file for patents could all play a role in explaining the greater patent relevance of US-American publications compared to EU publications.

As system integrators, large technology companies combine the entire process from R&D to commercialization under one roof. This makes it easier for them to transfer innovative ideas into applications.



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Fig. B2-10 Number of scientific publications in the field of quantum computing 2000–2022 and percentage cited in patents up to and including 2023



Legend: Authors with a German organizational affiliation were involved in 3,479 publications in the field of quantum computing between 2000 and 2022. By the end of 2023, 11.2 percent of these publications were cited in at least one patent.
Source: RelianceOnScience/Scopus/OpenAlex, own calculations.
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In the USA, for instance, Google and IBM not only invest large sums in basic research through their internal structures, but also systematically explore application potential and patent innovations. In Germany, on the other hand, the various steps in the R&I process are carried out by different players.

To ensure that excellent research is not conducted in Germany and the EU without a specific contribution to problem solving and commercial marketing, start-ups should play a key role in quantum computing. Start-ups develop innovative applications and bring them to market. They often emerge as spin-offs from research institutions and, due to their proximity to research institutions and specialized knowledge, serve as an important link between research and application. However, start-ups in Germany face structural challenges in both the start-up and growth phases. In particular, limited access to venture capital compared to the USA is a major problem for start-ups because they have both high capital requirements and a substantial risk of failure. In addition, the pool of potential investors is limited because they need specific expertise to

assess the potential of quantum projects. So far, private investment in German quantum computing start-ups has been very low compared to government funding.³¹⁴ A lack of follow-on financing, especially between the early seed phase (e.g. through start-up capital, funding programmes and angel investors) and later growth phases, often prevents start-ups in Germany from developing well.

B2-7 Recommendations for Action

Quantum technologies are potential key enabling technologies of the future, the research and development of which involves high costs and uncertainties. To ensure that German companies remain globally competitive with China and the USA and that Germany is technologically sovereign and well prepared for security policy threats in the long term, the Federal Government should promptly launch a comprehensive quantum strategy and play a major role in developing an efficient European quantum ecosystem. To this end, the Commission of Experts recommends:

Developing a National Quantum Strategy Rooted in a European Framework

- The Commission of Experts regards appropriate governance an urgent requirement for the development of a coherent strategic framework for current and future key enabling technologies such as quantum technologies. Accordingly, overarching coordination and strategic control should be handled centrally by the Federal Chancellery.³¹⁵
- In view of limited resources and international competition from the USA and China, an isolated national strategy is inefficient. Instead, national strategies within the EU should be more closely coordinated, synergies exploited and joint resources mobilized to ensure that EU companies are able to compete globally in R&I. The objective should be an efficient European quantum ecosystem.
- Germany and the EU should get involved in the development of uniform standards for safety assessment and certification at an early stage and take a leading role in the relevant committees. The aim should be to work towards internationally and globally valid standards that ensure that German and European interests are represented. Cooperation with institutions such as the US NIST could also be considered.
- German and European quantum strategies should aim to create good conditions for a quantum web that connects quantum technologies. To achieve this, the various quantum technologies must be thought of in an integrated way from the outset and their compatibility, e.g. at the software and language level, must be ensured.
- Since it is neither sensible nor realistic to assume that all raw materials and components for quantum computing will come from Germany or the EU in the long term, measures should be taken at an early stage to avoid unilateral dependencies on individual non-European companies or regions. While companies should primarily analyze their supply chains, diversify suppliers and develop strategic 'bargaining chips' for this purpose, it is the responsibility of political decision-makers in Germany and the EU to pursue a strategic foreign trade policy.

Updating the Quantum Strategy and Adapting it to the Technology Fields

- The funding framework of the Federal Government's current quantum strategy ends in 2026. To justify long-term investments in ongoing and future-orientated projects, the Federal Government should urgently develop an ambitious follow-up concept and swiftly implement it. The various fields of quantum technology differ in their potential areas of application and the demands they place on R&I policy. The follow-up concept should therefore also include specific measures for each field of technology and set clearly defined milestones in the form of a roadmap.
- In the field of classical computing, Europe is currently largely dependent on non-European companies. In quantum computing, however, Europe has the opportunity to develop and establish globally competitive technologies and companies. National competences and efforts must therefore be pooled across Europe to enable quantum computing 'Made in Europe' and to avoid dependencies on large non-European companies in the medium and long term. This requires a long-term technology roadmap that can be dynamically adapted as new insights emerge.³¹⁶
- In quantum communication, rapid and decisive action is needed to develop and implement new technologies to protect against interception and to ensure data integrity. To this end, activities should be coordinated and bundled throughout Germany and successful activities should be secured in the long term.
- In quantum sensing, Germany already has an excellent starting position thanks to excellent basic research and a broad corporate and user base. In contrast to other fields of technology, this has so far been maintained despite new technological developments. To secure a leading international position in the long term, the Federal Government should be committed to long-term funding of basic research in this field.

Expanding Regional R&I Clusters and Promoting Transregional Networking

- Existing regional R&I clusters for quantum technologies should be enabled to develop further with clear priorities and objectives, and to integrate research institutions, companies and investors in the process.
- The networking of R&I clusters should be actively supported to identify synergies in research, promote the exchange of knowledge between clusters and openness to alternative technological approaches. Care should be taken to ensure that a wide range of players have easy access to the research and computing infrastructure of the respective clusters across the regions.

Ensuring Transfer to Application and Exploiting Synergy Potential

- The Federal Government should increasingly consider concluding anchor customer contracts with companies in the field of quantum technologies for specified, but technology-neutral contracts with clear termination criteria. An anchor customer contract should be preceded by a competitive process in which suppliers develop new products in several stages in competition.
- A joint quantum benchmarking platform should be established to promote technology transfer and optimize feedback loops between R&D and applications. This should make it possible to evaluate the specific advantages of individual quantum technologies and technological solutions using comprehensive, clearly defined and generally accepted criteria.³¹⁷ A strong focus should be placed on the application benefits of the technologies.
- More consideration should be given to synergies between military and civilian use of

quantum technologies with regard to potential economies of scale and scope, as well as spillover effects, e.g. for encryption and navigation.

- Start-ups are vital for the transfer of quantum science findings into applications and are often spin-offs from research. A supportive environment with little red tape should be created to motivate researchers to turn their findings into applications. In addition, measures such as simplifying IP transfer and creating flexible career paths should be implemented to promote start-up activities.
- Sufficient capital is needed for the growth phase of young companies. This need is particularly pronounced in the field of quantum technology. The Federal Government should therefore take further measures to strengthen the venture capital market, for example by providing tax incentives and setting up investment funds. These approaches could mobilize private capital, facilitate the scaling of quantum companies and contribute to an attractive location for entrepreneurial activity.

Building Expertise and Increasing the Location's Attractiveness for Skilled Workers

- The Commission of Experts welcomes the programmes that are in place at the federal and Länder level to secure a supply of skilled workers in the field of quantum technologies. These programmes should be supplemented by additional measures, such as the introduction of specialized degree programmes, specific study modules and practice-orientated continuing education programmes.
- The Federal Government should take targeted measures to create an attractive environment for internationally mobile STEM graduates and quantum experts. This includes, not least, the swift processing of visa applications and the issuing of work and residence permits.³¹⁸